

Vibe Coding Chat History

PES1UG24AM271 — Lab 4: Lunar Lander

Scope: This document records the Vibe Coding interactions used to fix the broken code and implement the four features requested by the Lunar Lander repository README. Unrelated environment setup, Git troubleshooting, and video-recording discussion are intentionally excluded.

Task 1 — Fix touchdown / landing validation

Vibe Coding prompt

Fix the Lunar Lander landing validation. The current touchdown check only validates vertical speed and angle. Update it so that a landing is successful only when the horizontal speed is also within the repository's **MAX_SPEED_X** limit. Keep the existing vertical-speed and angle checks and preserve the existing crash behaviour.

ChatGPT implementation response

The landing condition was updated to check horizontal speed in addition to vertical speed and angle. The implementation introduced separate boolean checks for X speed, Y speed, and angle, and the angle-failure branch was updated to use the same validated angle condition.

Relevant implementation excerpt

```
ok_x = abs(self.vel.x) <= MAX_SPEED_X
ok_y = abs(self.vel.y) <= MAX_SPEED_Y
ok_a = abs(angle) <= MAX_ANGLE

if pad and ok_x and ok_y and ok_a:
```

Testing / outcome

The game was run and the landing logic was tested successfully. Commit created: **967980c — Fix horizontal landing speed.**

Task 2 — Add fuel-based ship colour

Vibe Coding prompt

Add fuel-based ship colour to the Lunar Lander. The ship should be green when fuel is high, yellow when fuel is medium, and red when fuel is low. Make the thresholds clear and ensure the fuel ratio stays within the valid 0–1 range.

ChatGPT implementation response

A helper function was added to convert the current fuel ratio into a colour. The ratio is clamped between 0 and 1, then mapped to green, yellow, or red according to the fuel level.

Relevant implementation excerpt

```
def ship_color(fuel_ratio):  
    ratio = max(0.0, min(1.0, fuel_ratio))  
    if ratio > 0.6:  
        return (80, 220, 90)  
    if ratio > 0.2:  
        return (220, 200, 50)  
    return (220, 70, 70)
```

Testing / outcome

The updated game was run and the colour feature was verified. Commit created: **1308f4c — Add fuel-based ship color**.

Task 3 — Add landing fireworks / particle effect

Vibe Coding prompt

Add a fireworks or particle effect that appears after a successful landing. It should be visually noticeable, use multiple particles/colours, and integrate with the existing Pygame update and drawing loop without breaking the game.

ChatGPT implementation response

A landing-fireworks particle system was added. The implementation stores active particles, spawns them at the successful landing position, updates their motion over time, and draws them with multiple colours. The landing position is retained so the effect can be triggered after a successful landing.

Relevant implementation excerpt

```
LAST_LANDING_POS = None
FIREWORKS = []

def _spawn_fireworks(pos):
    ...
    FIREWORKS.append(...)
```

Testing / outcome

The game was run, a successful landing was performed, and the fireworks effect was observed. Commit created: **ac91eef — Add landing fireworks.**

Task 4 — Bonus life every 1500 points

Vibe Coding prompt

Add a bonus life system. The player should receive an extra life whenever the score reaches each 1500-point threshold: 1500, 3000, 4500, and so on. The same threshold must not award multiple lives.

ChatGPT implementation response

The bonus-life threshold was implemented around a 1500-point interval. The existing score/life handling advances through successive thresholds so that each threshold is awarded once rather than repeatedly for the same score range.

Relevant implementation excerpt

```
def bonus_life_threshold():  
    return 1500
```

Testing / outcome

The game was run after implementing the bonus-life feature. Commit created: **63f6245 — Add bonus life at 1500 points.**

Final Vibe Coding Summary

Task	Feature	Commit
1	Horizontal landing-speed validation	967980c
2	Fuel-based ship colour	1308f4c
3	Landing fireworks	ac91eef
4	Bonus life every 1500 points	63f6245

Note: This is a focused Vibe Coding record reconstructed from the task-specific interaction. It is not a verbatim export of unrelated setup, Git, video, or submission troubleshooting.